Self-monitoring of blood glucose by diabetic women during the third trimester of pregnancy

Ulf Hanson, M.D., Bengt Persson, M.D., Erik Enochsson, M.D., Per Lennerhagen, M.D., Fredrik Lindgren, M.D., Viveca Lundström, M.D., Nils-Olov Lunell, M.D., Bo A. Nilsson, M.D., Lennart Nilsson, M.D., Magnus Stangenberg, M.D., Bertil Thalme, M.D., Karl-Gunnar Tillinger, M.D., Ulrika Öfverholm, M.D.

Stockholm, Sweden

The clinical value of self-monitoring blood glucose in diabetic pregnancy at home was compared with hospital care in the thirty-second to thirty-sixth week of pregnancy in a prospective randomized study including 100 pregnancies in 97 patients (White's class B, 38; C, 25; D, 28; and F, 9) of which 54 were in the home group and 46 in the hospital group. The duration of pregnancy was not significantly different in two treatment groups, with a median duration of 266.0 days in the home group and 266.5 days in the hospital group. The mean blood glucose values during the study period were 5.9 mmol/L in the home group and 6.0 mmol/L in the hospital group, thus there were no significant group differences. There were no significant group differences in pregnancy complications; however, 10 of 54 (19%) had to interrupt home-monitoring because of pregnancy complications. The perinatal morbidity was not significantly different in the two treatment groups, with the following percentages of complications in the combined series: 4% idiopathic respiratory distress syndrome, 7% transient tachypnea, 2% symptomatic hypoglycemia, 16% hyperbilirubinemia, 22% feeding problems, and 10% erythrocytosis.
(AM J OBSTET GYNECOL 1984;150:817-21.)

To achieve adequate blood glucose control in diabetic women during pregnancy, frequent hospitalization has often been used, especially in the third trimester. The introduction of techniques for the self-monitoring of blood glucose has made it possible to reduce the periods of hospitalization. Pregnant diabetic patients have an increased incidence of complications, such as hypertension and premature labor. Control of the patient on an outpatient basis instead of in a hospital ward in the third trimester of pregnancy may influence the frequency of such complications.

Although most of the published data clearly demonstrate the feasibility of self-monitoring of blood glucose during pregnancy,[1-4] the published series are either too small or do not include control groups that would allow an evaluation of its clinical value in terms of pregnancy complications and perinatal outcome.

The aim of this prospective randomized study was to compare the clinical value of home blood glucose monitoring in diabetic pregnancies with traditional hospital care from the thirty-second to the thirty-sixth week of pregnancy.

From the Departments of Obstetrics and Gynecology and Pediatrics, Karolinska Hospital, Huddinge Hospital, Danderyd Hospital, Söderhospital–Sachsska Infants Hospital, Sabbatsberg Hospital, and St. Görans Hospital.
Supported by grants from Expressens Perinatalforskningsfond, Allmänna Barnbördshusets Minnesfond, Svenska Diabetesstiftelsen, Nordisk Insulinfond, Swedish Medical Research Council (Project No. 3787), and Tielman's Fund for Pediatric Research.
Received for publication December 1, 1983; revised June 1, 1984; accepted June 11, 1984.
Reprint requests: Dr. Ulf Hanson, Department of Obstetrics and Gynecology, Karolinska Hospital, S-104 01 Stockholm, Sweden.

Material and methods

Procedure. The study started on October 1, 1979, and the last patient was delivered before October 1, 1982. In the Stockholm area five hospitals administer care of diabetic pregnancies, and all of them participated in the study. Patients with a diagnosis of diabetes, either insulin-dependent or non-insulin-dependent prior to pregnancy, were in each clinic randomized into two treatment groups: (1) the hospital group, including patients who were hospitalized from the thirty-second week until delivery and (2) the home group, including patients who self-monitored their blood glucose at home from the thirty-second week until the thirty-sixth week of gestation and then were hospitalized during the thirty-seventh week until delivery.

The study was approved by the Regional Ethical Committee. At the time of the first examination during the pregnancy, the patients were informed about the study and consented to participate.

Patients. During the study period there was a total of 110 pregnancies in 107 diabetic patients. Only one patient could not participate in the study. She was severely addicted to drugs and had a severe angiopathy

Table I. Distribution of patients with a pregnancy length of ≥28 prior to each pregnancy according to White classification*

	White Class				
	B	*C*	*D*	*F*	*Total*
Hospital group	15	13	15	3	46
Home group	23†	12	13	6	54
Total	38	25	28	9	100

*There were no significant differences between the two treatment groups.

†Including two twin pregnancies.

(White Class F). There were eight spontaneous abortions in the remaining 109 pregnancies. One mother (White Class F) died, probably because of hypoglycemia, in her home during the thirteenth week of pregnancy. A total of 100 pregnancies in 97 patients continued at least as far as the twenty-eighth week of pregnancy. Among these 97 patients, 83 were insulin dependent and 14 were not. All of the 14 non-insulin-dependent patients were treated with insulin from the beginning of the pregnancy. No gestational diabetic patients were included. There were two twin pregnancies during this period and 98 singleton pregnancies.

Blood glucose and hemoglobin A_{1c} determinations. Blood glucose was determined four times daily (7 AM, 9:30 AM, 3 PM, 7 PM) from the thirty-second week of gestation until delivery in both groups. Blood glucose was determined by a glucose-oxidase method,[5] or with a reflectometer (Reflomat) in the hospital and in the home group, respectively. Samples for hemoglobin A_{1c} determinations were taken at the first examination and then during the thirty-second and the thirty-sixth weeks of pregnancy and around the time of delivery. The hemoglobin A_{1c} value was determined by isoelectric focusing.[6]

Pregnancy supervision. Both groups were given the same dietary instructions. The prescribed diet consisted of 50% of calories from carbohydrate sources, 20% from protein, and 30% from fat. The average prescribed daily caloric intake was 1800 kcal during the third trimester and the mean weight gain during pregnancy was 12.7 ± 4.2 kg (mean ± S.D.) for the combined two series. Until the thirty-second week of pregnancy, the patients were seen at least every other week in the outpatient clinic. From the thirty-second week, the home group was seen weekly in the antenatal clinic. In both groups fetal growth was assessed by measuring the symphyseal-fundal height and the biparietal diameter. The fetal condition was also assessed, usually by a weekly nonstress test, and the placental function was evaluated by determining 24-hour estriol content at least once a week. If complications occurred, the home-monitoring situation was interrupted. Hypertension in pregnancy was diagnosed if the blood pressure was ≥140/90 mm Hg. Preeclampsia was defined as hypertension in pregnancy combined with proteinuria. In patients with preexisting proteinuria (White Class F) a blood pressure reading of ≥140/90 mm Hg combined with at least a 100% increase in proteinuria served as indication of preeclampsia. Premature delivery was defined as delivery before the end of the thirty-seventh week of pregnancy, i.e., a gestational age of <259 days according to the World Health Organization.

Neonatal care. All infants were routinely observed in a neonatal care unit for 1 to 3 days. Early feeding was practiced.[7] Blood samples for analyses of hemoglobin, hematocrit, and glucose were taken at about 2 hours of age. Blood glucose was thereafter monitored every 4 hours, at least during the first 24 hours after birth. Serum calcium and bilirubin levels were determined, if required. Feeding problems were considered to be present if the early feeding program could not be followed or if the infant required feeding by the nasogastric route and/or had signs of a significant amount of gastric residuum.

Respiratory disorders were classified according to the criteria stated by Hjalmarsson.[8] A hematocrit value >70% was regarded as erythrocytosis. Hypoglycemia was defined as a blood glucose <1.7 mmol/L. Symptomatic hypoglycemia was defined as relief of symptoms of hypoglycemia by the administration of intravenous glucose. Hyperbilirubinemia was defined as a serum bilirubin value of ≥300 μmol/L.

Statistical methods. The tables and figures present data from the total series of 100 pregnancies: 54 in the home group and 46 in the hospital group. However, this material includes two pairs of twins, and three mothers had two pregnancies during the study period. The two twin pregnancies and the second pregnancy of the three mothers with two pregnancies each were excluded from the statistical calculations in order to have independent samples. Furthermore, when comparing by analysis of variance the blood glucose weekly mean values and hemoglobin A_{1c} values during the study period (the thirty-second to the thirty-sixth weeks of pregnancy), those patients who were delivered before the thirty-sixth week, those who because of pregnancy complications had to interrupt the home-monitoring, and those patients in the hospital group with the same type of complications had to be excluded in order to have statistically comparable groups. Blood glucose values and hemoglobin A_{1c} values were log-transformed before calculations in order to obtain normal distributed values. The given means and 95% confidence intervals

Table II. Maternal complications and neonatal outcome in two treatment groups*

	Hospital group		*Home group*		*Total*	
	n	%	*n*	%	*n*	%
Maternal complications						
No. of pregnancies	46	100	54	100	100	100
Hypertension in pregnancy	4	9	2	4	6	6
Preeclampsia	1	2	5	9	6	6
Placenta praevia	1	2	0	0	1	1
Abruptio placenta	1	2	2	4	3	3
Pulmonary embolism	0	0	1	2	1	1
Premature delivery						
Induced	9	20	9	17	18	18
Spontaneous	5	11	5	9	10	10
Neonatal outcome†						
No. of infants	46	100	56‡	100	102‡	100
Major congenital malformations	3	7	1	2	4	4
Respiratory distress syndrome	1	2	3	5	4	4
Transient tachypnea	4	9	3	5	7	7
Hypoglycemia, total	11	24	13	23	24	24
Hypoglycemia, symptomatic	0	0	2	4	2	2
Hyperbilirubinemia	3	7	8	14	11	11
Feeding problems	11	24	11	20	22	22
Erythrocytosis	5	11	5	9	10	10

*There were no significant differences between the two treatment groups.
†Median birth weights were 3395 gm for the hospital group and 3260 gm for the home group (NS).
‡Two pair of twins are included.

in Figs. 1 and 2 are the antilogged values. Only those patients with a pregnancy length of ≥32 weeks were included in group comparisons. Group distributions were analyzed by the χ^2-test or Fischer's test. Mean, median, and confidence intervals were calculated by conventional methods. Group differences were calculated by the Mann-Whitney test. A p value of <0.05 was regarded as a significant difference.

Results

White classification. The distribution of the patients prior to each of the 100 pregnancies by use of the White classification is given in Table I. There were no significant differences between the distributions in the two groups.

Pregnancy complications. The number and type of various complications during pregnancy (Table II) were not significantly different between the two groups. There were 28 deliveries before 37 completed weeks of gestation, and 18 of these preterm deliveries were induced mainly because of pregnancy complications. The duration of pregnancy was not significantly different between the two groups. The median duration was 266.5 days in the hospital group and 266.0 days in the home group. Ten of 54 patients (19%) in the home group had to be hospitalized because of pregnancy complications; in other words, the self-monitoring situation was interrupted. Three patients in each treatment group were delivered because of suspicion of abnormal fetoplacental function (low urine estriol values or abnormal nonstress test). There were no significant differences in the route of delivery in the two groups. The cesarean section rates were 48% in the home group and 50% in the hospital group.

Blood glucose and hemoglobin A_{1c} values. Weekly mean blood glucose values were calculated from the four daily values. The group weekly mean values and the 95% confidence intervals are shown in Fig. 1. The mean blood glucose values during the study period were 6.0 mmol/L (108 mg%) for the hospital group and 5.9 mmol/L (106 mg%) for the home group. The mean hemoglobin A_{1c} values and the 95% confidence intervals from different times during pregnancy are shown in Fig. 2. There were no significant differences between the two treatment groups in hemoglobin A_{1c} or blood glucose values during the study period.

Hospital care. The median duration of prelabor hospital care in the third trimester was 39 days (range, 1 to 70 days) in the hospital group and 20 days (range, 0 to 51 days) in the home group. The difference is significant (Mann-Whitney $p < 0.001$).

Perinatal mortality. There was one perinatal death in each group i.e., a perinatal mortality of 2%. One infant was delivered at 28 weeks' gestation because of placenta previa and hemorrhage. The infant died immediately after delivery of pulmonary and renal hypoplasia and a severe fetomaternal transfusion syndrome. The other infant's mother had severe pro-

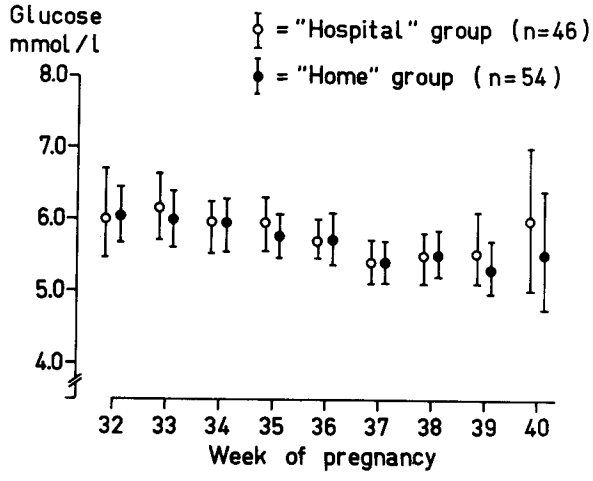

Fig. 1. Weekly mean values and 95% confidence intervals of blood glucose in the two treatment groups. No significant differences were noted between the two groups during the study (analysis of variance, not significant) (1 mmol/L = 18.0 mg/100 ml).

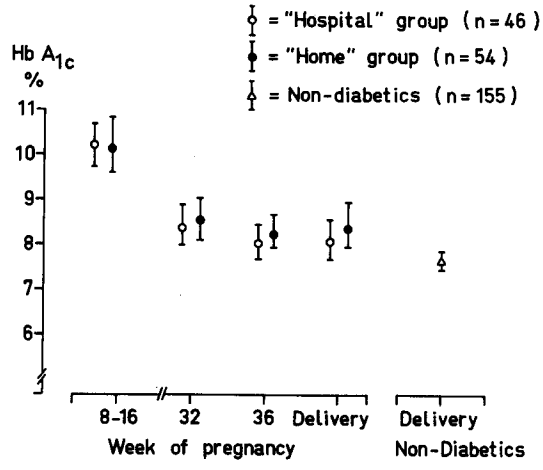

Fig. 2. Hemoglobin A_{1c} mean values and 95% confidence intervals at different times during pregnancy. No significant differences were noted between the two treatment groups during the study (analysis of variance, not significant).

liferative retinopathy and nephropathy. The delivery occurred in the thirtieth week because the mother had a rapidly progressing preeclampsia. The infant died of idiopathic respiratory distress syndrome 2 days later. This gives a corrected total perinatal mortality of 1%.

Neonatal outcome. The frequency of neonatal complications, such as congenital malformations, respiratory disorders, hypoglycemia, hyperbilirubinemia, erythrocytosis, and feeding problems are given in Table II. There were no significant differences between the two groups. No significant difference in the median duration of stay in the neonatal unit was noted between the two groups, the median time being 2.7 days in the home group and 2.8 days in the hospital group.

Comments

This prospective randomized study demonstrates that it is possible to achieve the same degree of blood glucose control and the same hemoglobin A_{1c} levels by self-monitoring of blood glucose at home as by hospital care in diabetic patients during the last trimester of pregnancy (weeks 32 to 36). This can be achieved without any significant increase in the frequency of pregnancy complications or perinatal mortality and morbidity. In this study all pregnant patients with diabetes but one (a case of severe drug addiction) were included from a defined geographical area during a 3-year period. Thus, since no other criteria for selection of the patients were applied, the results indicate that it is feasible to use self-monitoring of blood glucose in the routine management of an unselected group of pregnant diabetic patients in the third trimester. One drawback in a multicenter study could be the minor differences in the management of patients between various hospitals. These difficulties were offset by the fact that the patients at each hospital were randomly distributed to the two treatment groups.

An important aim of the study was to assess the effect of self-monitoring on the occurrence of pregnancy complications, such as toxemia and premature labor. There were no significant differences between the two groups with regard to these complications. However, 10 patients (19%) in the home group had to interrupt the home-monitoring situation because of pregnancy complications. This underlines the importance of well-organized and qualified outpatient obstetric supervision of pregnant diabetic patients, especially if the management program includes self-monitoring of blood glucose at home during the third trimester.

In summary, our data suggest that the frequency of pregnancy complications, the blood glucose control, and the perinatal mortality and morbidity rates are not significantly different between mothers who have monitored their blood glucose at home and mothers who have been treated in hospital during the same period in the third trimester of pregnancy.

We are grateful to Mrs. Gun-Marie Taube for assistance in taking care of all the clinical data and to Mrs. Astrid Häggblad for preparing the manuscript.

REFERENCES

1. Peacock I, Hunter IC, Walford S, et al. Self-monitoring of blood glucose in diabetic pregnancy. Br Med J 1979;2:1333.
2. Irsigler K, Boli-Tanbald C. Self-monitored blood glucose: the essential signal in the diabetic patient's effort to achieve normoglycemia. Diabetes Care 1980;3:163.

3. Skyler IS, O'Sullivan MI, Robertson EG, et al. Blood glucose control during pregnancy. Diabetes Care 1980;3:69.
4. Jovanovic L, Peterson CM, Saxena BB, et al. Feasibility of maintaining normal glucose profiles in insulin-dependent pregnant diabetic women. Am J Med 1980;68:105.
5. Trinder A. Determination of glucose in blood using glucose oxidase with an oxygen acceptor. Ann Clin Biochem 1969;6:24.
6. Jeppsson JO, Franzén B, Gaal AB. Simplified determination of hemoglobin A_{1c} in diabetic patients by use of electrofocusing. In: Rodala BJ, ed. Electrophoresis: advanced methods in biochemical and clinical applications. Berlin: Gruyter de Walter, 1980:655.
7. Persson B, Gentz I, Feychting H. Management of the infant of the diabetic mother. In: Sutherland HW, Stowers IM, ed. Carbohydrate metabolism in pregnancy and the newborn, Edinburgh: Churchill Livingstone, 1975:232.
8. Hjalmarsson O. Epidemiology and classification of acute, neonatal respiratory disorders: a prospective study. Acta Pediatr Scand 1981;70:773.

Rise in serum levels of PZ-peptidase, an enzyme involved in collagen breakdown, in human pregnancy and labor

Mohammad Rajabi, M.D., and J. Frederick Woessner, Jr., Ph.D.
Miami, Florida

PZ-Peptidase is an enzyme capable of cleaving small peptides that are produced during the metabolic breakdown of collagen. It is conveniently assayed by use of the synthetic peptide 4-phenylazobenzyloxycarbonyl-L-prolyl-L-leucyl-glycyl-L-prolyl-D-arginine. An improved assay is reported which permits measurements of small amounts of this enzyme in 0.1 ml of serum. The enzyme activity in serum was found to rise from 1.3 U/L (1 U = 1 μmol cleaved per minute) in nongravid patients to 4.0 U/L in patients at term, who display no objective or subjective signs of labor. Patients in active labor show a further elevation of activity to 6.5 U/L. These results are believed to reflect changes in collagen metabolism related to remodeling of the uterus and skeleton in pregnancy and to cervical dilatation and placental separation at the time of labor. (AM J OBSTET GYNECOL 1984;150:821-6.)

The present study examines human serum in an effort to detect enzymes that might be associated with the process of cervical dilatation. The human cervix is largely a connective tissue structure, with less than 15% smooth muscle content.[1] Cervical dilatation involves changes both in the collagen fibers[2, 3] and in the glycoprotein and proteoglycan components of the extracellular matrix.[2, 4] The collagen fibers are somehow altered so that the tensile properties of the cervix are greatly reduced. This involves increased solubilization of collagen[3, 5] and a 50% to 70% reduction in collagen per unit weight by term.[6, 7]

If degradative alterations of the collagen fibers are important in dilatation, as is indicated by histologic studies,[6, 8, 9] then one must think of proteolytic enzymes that might be involved. Collagen requires a specific collagenase to cleave the helical region of the molecule. There is some evidence that collagenase is elevated in the cervix at term.[10] After the helix is cleaved, the collagen fragments denature to gelatin at body temperature. Gelatinase action would cleave these fragments to small peptides, which could then be further cleaved by DNP-peptidase and PZ-peptidase. These two peptidases are also known to be elevated in the cervix at term.[10–12]

Since collagen is an extracellular protein, these proteolytic enzymes involved in its destruction may act outside the cells and may spill over into the bloodstream. Collagenase and gelatinase would probably be blocked by serum proteinase inhibitors, especially by α_2-macroglobulin.[13] PZ-Peptidase, on the other hand, is not known to react with this inhibitor; even if it did, the small substrate could still penetrate the enzyme-inhibitor complex and be cleaved at a rapid rate. Since a method had previously been published by Gries et al.[14] for the measurement of this enzyme in serum, PZ-peptidase was selected for a study of its changes during pregnancy and parturition. In most studies of the cervix, measurements have been made on nongravid patients and then on patients at term or imme-

From the Division of Maternal-Fetal Medicine, Department of Obstetrics and Gynecology and Department of Biochemistry, University of Miami School of Medicine.
Supported by Grant HD-06773 from the National Institute of Child Health and Human Development.
Received for publication March 14, 1984; revised April 20, 1984; accepted June 4, 1984.
Reprint requests: Dr. J. F. Woessner, Jr., Department of Biochemistry R-127, P. O. Box 016960, Miami, FL 33101.